Small Signal Schottky Diode

FEATURES

- For general purpose applications
- This diode features low turn-on voltage
- The devices are protected by a PN junction guard ring against excessive voltage, such as electrostatic discharges
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT

HALOGEN FREE

LINKS TO ADDITIONAL RESOURCES

3D Models | Marking | Parametric Search | Order Samples

APPLICATIONS

- Applications where a very low forward voltage is required

MECHANICAL DATA

Case: MiniMELF (SOD-80)

Weight: approx. 31 mg

Cathode band color: black

Packaging codes/options:

18/10K per 13" reel (8 mm tape), 10K/box

08/2.5K per 7" reel (8 mm tape), 12.5K/box

PARTS TABLE

PART	ORDERING CODE	CIRCUIT CONFIGURATION	REMARKS
BAS85-M	BAS85-M-18 or BAS85-M-08	Single	Tape and reel

ABSOLUTE MAXIMUM RATINGS (T_{amb} = 25 °C, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Continuous reverse voltage		V_R	30	V
Forward continuous current (1)		I_F	200	mA
Peak forward current (1)		I_{FM}	300	mA
Surge forward current (1)	t_p < 1 s	I_{FSM}	600	mA
Power dissipation (1)	T_{amb} = 65 °C	P_{tot}	200	mW

Note

(1) Valid provided that electrodes are kept at ambient temperature.

THERMAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

PARAMETER	TEST CONDITION	SYMBOL	VALUE	UNIT
Thermal resistance junction to ambient air (1)		R_{thJA}	430	K/W
Junction temperature		T_j	125	°C
Storage temperature range		T_{stg}	-55 to +150	°C
Operating temperature range		T_{op}	-55 to +125	°C

Note

(1) Valid provided that electrodes are kept at ambient temperature.

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ELECTRICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)						
PARAMETER	**TEST CONDITION**	**SYMBOL**	**MIN.**	**TYP.**	**MAX.**	**UNIT**
Reverse breakdown voltage	I_R = 10 µA (pulsed)	$V_{(BR)}$	30			V
Leakage current	V_R = 25 V	I_R		0.2	2	µA
Forward voltage	Pulse test t_p < 300 µs, I_F = 0.1 mA	V_F			240	mV
	Pulse test t_p < 300 µs, I_F = 1 mA	V_F			320	mV
	Pulse test t_p < 300 µs, I_F = 10mA	V_F			400	mV
	Pulse test t_p < 300 µs, I_F = 30 mA	V_F		500		mV
	Pulse test t_p < 300 µs, I_F = 100 mA	V_F			800	mV
Diode capacitance	V_R = 1 V, f = 1 MHz	C_D			10	pF
Reverse recovery time	I_F = 10 mA, I_R = 10 mA, i_R = 1 mA	t_{rr}			5	ns

TYPICAL CHARACTERISTICS (T_{amb} = 25 °C, unless otherwise specified)

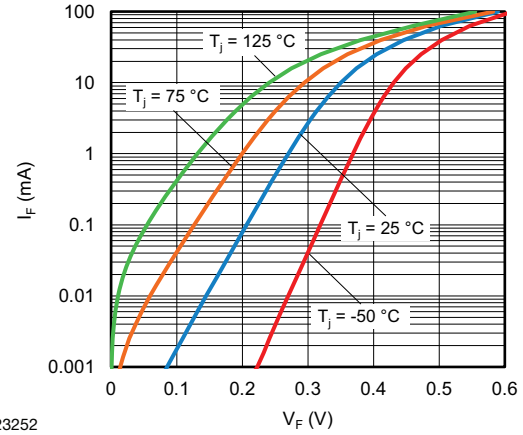

Fig. 1 - Typical Forward Current vs. Forward Voltage

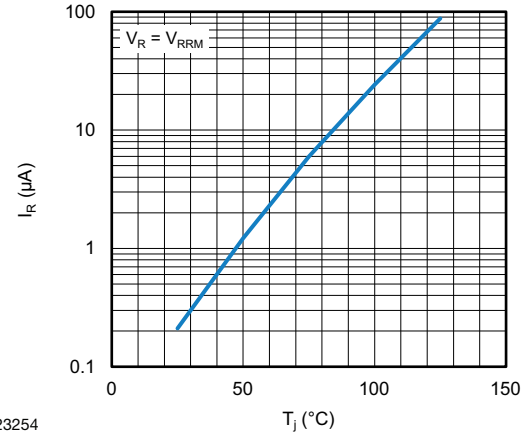

Fig. 3 - Typical Reverse Current vs. Junction Temperature

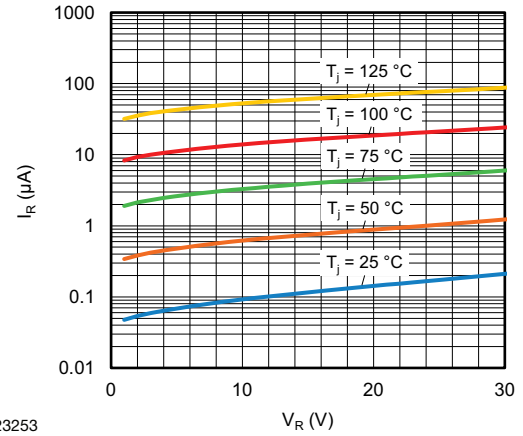

Fig. 2 - Typical Reverse Leakage Current vs. Reverse Voltage

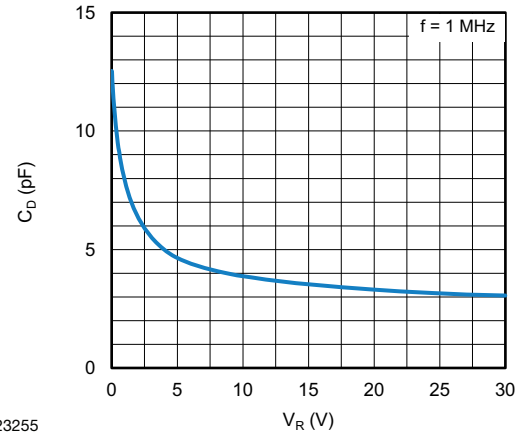

Fig. 4 - Typical Capacitance vs. Reverse Voltage

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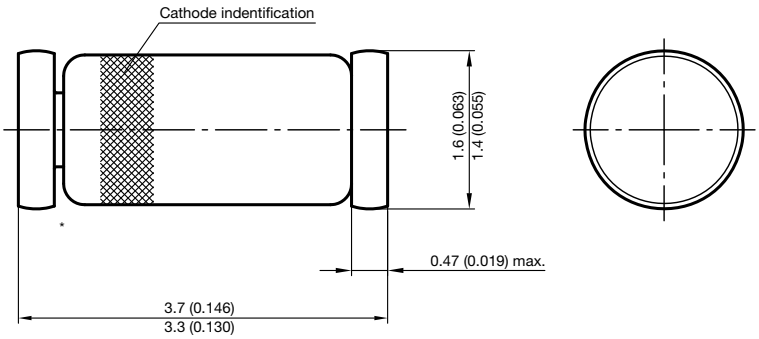
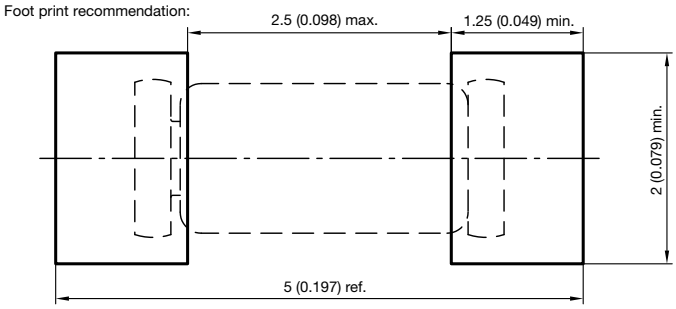
PACKAGE DIMENSIONS in millimeters (inches): **MiniMELF (SOD-80)**





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